



Machines Learn in Many Different Ways

This article gives the reader a bird's eye view of machine learning models, and solves a use case through Sframes and Python.

‘Data is the new oil’—and this is not an empty expression doing the rounds within the tech industry. Nowadays, the strength of a company is also measured by the amount of data it has. Facebook and Google offer their services free in lieu of the vast amount of data they get from their users. These companies analyse the data to extract useful information. For instance, Amazon keeps on suggesting products based on your buying trends, and Facebook always suggests friends and posts in which you might be interested. Data in the raw form is like crude oil—you need to refine crude oil to make petrol and diesel. Similarly, you need to process data to get useful insights and this is where machine learning comes handy.

Machine learning has different models such as regression, classification, clustering and similarity, matrix factorisation, deep learning, etc. In this article, I will briefly describe these models and also solve a use case using Python.

Linear regression: Linear regression is studied as a model to understand the relationship between input and output numerical values. The representation is a linear equation that combines a specific set of input values (x), the solution to which is the predicted output for that set of input values. It helps in estimating the values of the coefficients used in the representation with the data that we have available. For example, in a simple regression problem (a

single x and a single y), the form of the model is:

$$y = B_0 + B_1x$$

Using this model, the price of a house can be predicted based on the data available on nearby homes.

Classification model: The classification model helps identify the sentiments of a particular post. For example, a user review can be classified as positive or negative based on the words used in the comments. Given one or more inputs, a classification model will try to predict the value of one or more outcomes. Outcomes are labels that can be applied to a data set. Emails can be categorised as spam or not, based on these models.

Clustering and similarity: This model helps when we are trying to find similar objects. For example, if I am interested in reading articles about football, this model will search for documents with certain high-priority words and suggest articles about football. It will also find articles on Messi or Ronaldo as they are involved with football. TF-IDF (term frequency - inverse term frequency) is used to evaluate this model.

Deep learning: This is also known as deep structured learning or hierarchical learning. It is used for product recommendations and image comparison based on pixels.